

Smart Metro Ticket Booking System

IDEATION PHASE

Date

Team ID

Project Name: Smart Metro Ticket Booking System

Maximum Marks: 42 Marks

1. Problem Statement

1.1 Background

Among urban public transport options, the metro is one of the most heavily used, prized for its speed, affordability, and role in easing road traffic. Even so, buying a metro ticket in many networks still relies mainly on staffed counters, coin-operated token machines, or kiosks that are only partially digitised.

A rider typically has to go to the station in person, wait their turn in a line, pick a source and destination, hand over cash or a card, and walk away with a paper ticket or token before being let onto the platform.

1.2 Problem Statement

"Metro passengers regularly deal with long counter queues, fare mistakes from manual calculation, restricted counter hours, and reliance on paper tickets or tokens — all of which add delay, crowd the ticketing area, and make travel less pleasant, particularly during rush hour or for first-time and differently-abled riders."

1.3 Problem Statement Canvas (Who / What / Why)

Element	Description
I am (User)	An everyday commuter — office-goer, student, occasional rider, senior citizen, or a differently-abled passenger — who relies on the metro.
Trying to	Get a metro ticket booked fast for a chosen source and destination, skipping the wait in line.
But I am	Stuck waiting at busy ticket counters or dealing with vending machines that are sluggish, offline, or hard to use.
Because	Ticket issuing is mostly a manual job, counters are short-staffed at busy times, and fares are worked out by hand.

1.4 Key Points

- Ticket-counter lines stretch long during office rush, exam season, and festival travel.
- Working out fares by hand is slow and leaves room for mistakes.
- Physical counters keep limited hours even though travel demand runs all day.
- A lost or damaged token/paper ticket means extra time re-verifying at the counter.
- Elderly, disabled, and visually impaired riders get little dedicated help at manual counters.
- There's no single digital record of a rider's past trips or spending.
- Running counters, staffing them, and stocking tokens is costly for the metro authority.
- Fare collection isn't linked to any workflow-based support or refund process.

1.5 Impact of the Problem

1.6 Need for a Digital Solution

A self-service digital booking system is clearly needed — one where a rider picks their stations, sees the fare worked out automatically, and gets a QR-based digital ticket right away, with no counter visit involved. Delivering this on the ServiceNow platform means the solution can draw on the Service Catalog for a guided booking flow,

Flow Designer to automate fare calculation and ticket issuance, and Notifications to deliver the digital ticket instantly — producing a quicker, more inclusive, and more dependable travel experience.